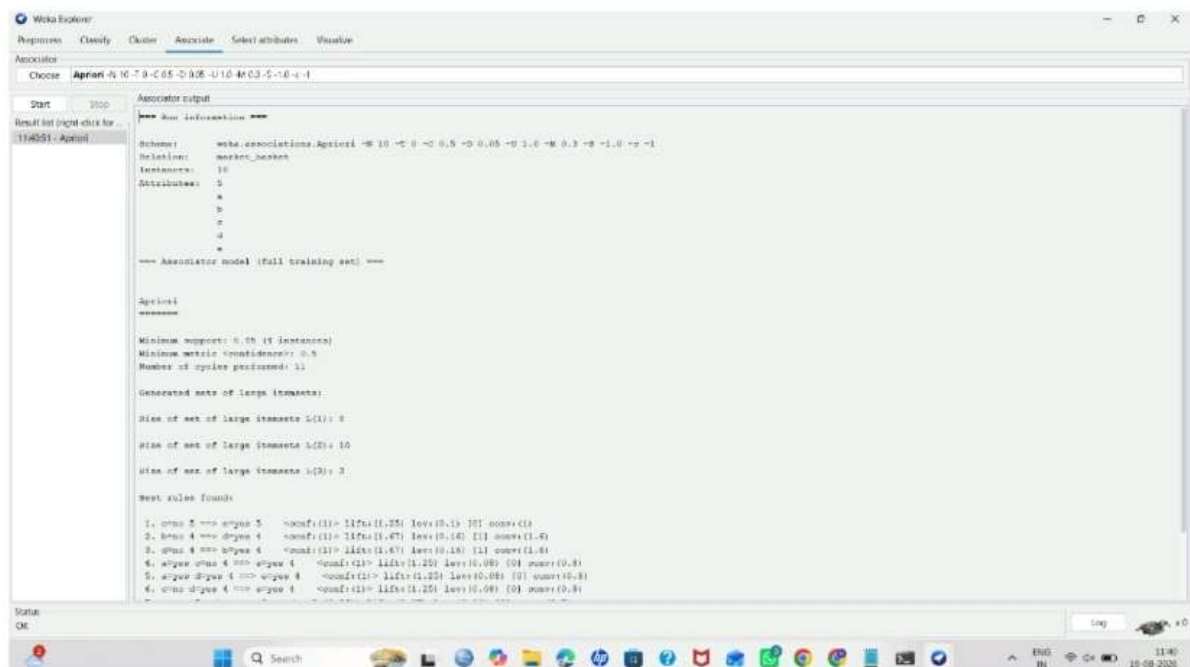
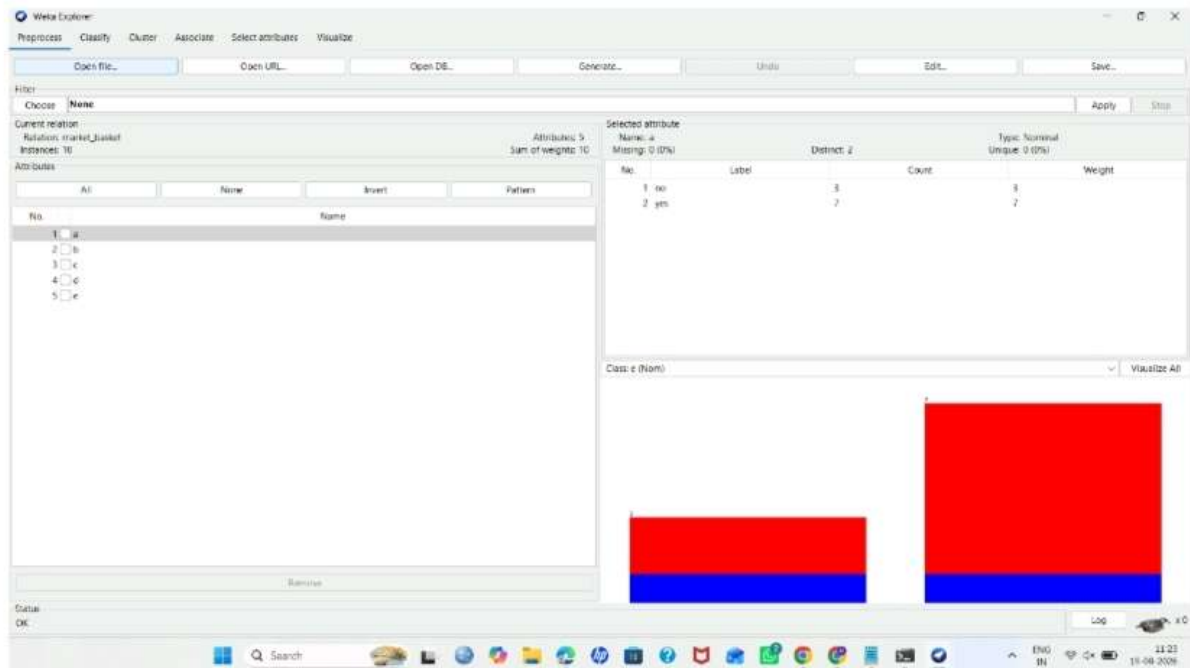


UNIT-3 LAB EXPERIMENTS

1.



2.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply Cancel

Current relation: Relations market_basket
Instances: 10
Attributes: 5
Sum of weights: 10

Attributes: All None Invert Pattern

No.	Name
1	Milk
2	Bread
3	Diapers
4	Bread
5	Butter
6	Cookies

Remove

Status: OK

Selected attribute: Name: Milk
Missing: 0 (0%)

No.	Label	Count	Type: Nominal	Unique: 5 (100%)	Weight
1	yes	5			
2	no	5			

Class Cookies (Nom)

Log

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associate: Choose **Apriori** $N=20$ $\alpha=0.05$ $\beta=0.05$ $\gamma=0.1$ $M=2$ $S=10$ $\epsilon=1$

Start Stop

Result not input-click for ...
11/07/2020 - Apriori

Associate output

```

*** Run Information ***
Scheme: weka.associations.Apriori -M 20 -C 0 -O 0.5 -D 0.05 -I 0.1 -M 2 -S 10 -E 1
Relation: market_basket
Instances: 10
Attributes: 6
  Milk
  Bread
  Diapers
  Butter
  Cookies

*** Associate model (full training set) ***

Apriori
=====

Minimum support: 0.05 (4 instances)
Minimum metric (confidence): 0.5
Number of cycles performed: 11

Generated sets of large itemsets:

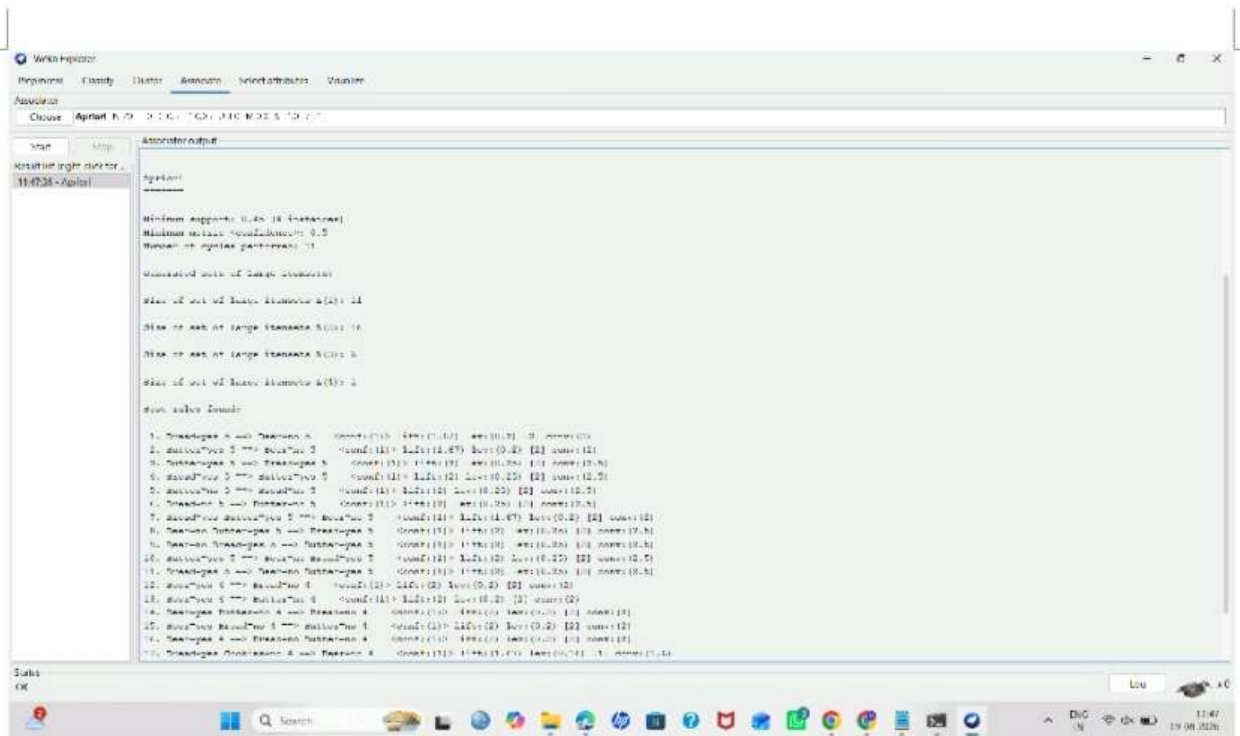
Size of set of large itemsets L(1): 11
Size of set of large itemsets L(2): 16
Size of set of large itemsets L(3): 4
Size of set of large itemsets L(4): 1

Best rules found:

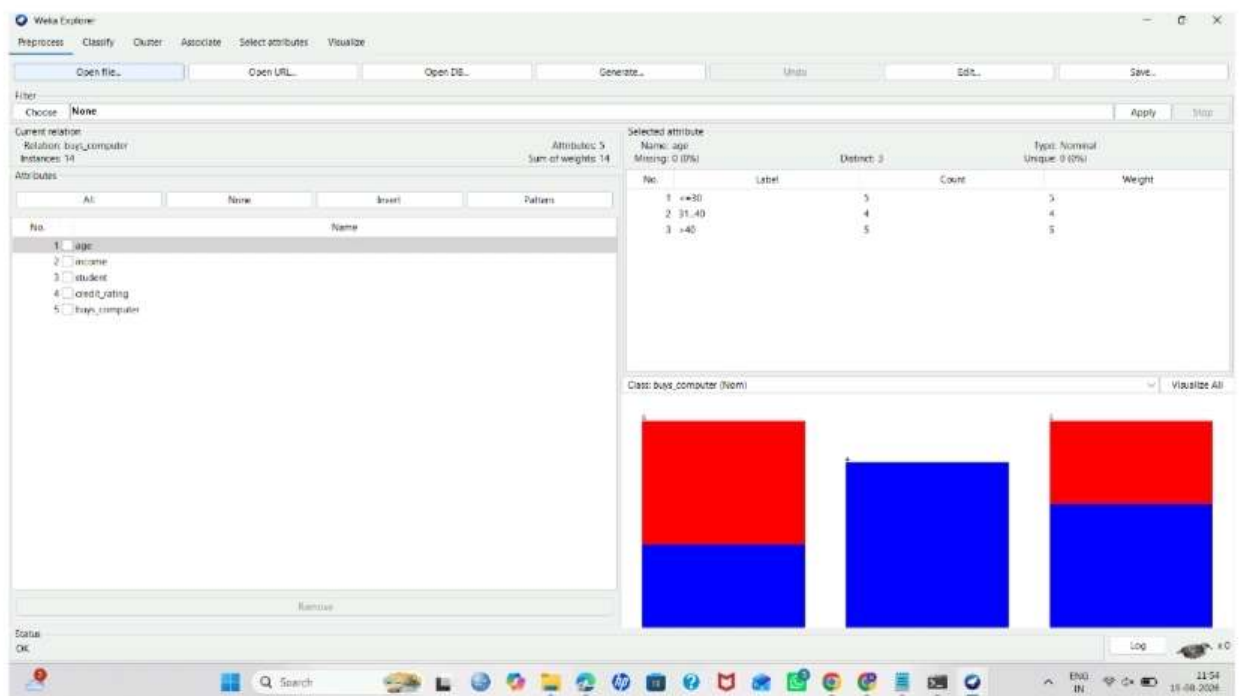
1. Bread=yes & Butter=no => Cookies=1  conf:(1) > 11Pr(1.67) lev:(0.2) [5] cover:(5)
2. Diapers=yes & Bread=no => Cookies=1  conf:(1) > 11Pr(1.67) lev:(0.2) [2] cover:(2)
3. Diapers=yes & Bread=yes => Cookies=1  conf:(1) > 11Pr(1.67) lev:(0.2) [2] cover:(2,3)
  
```

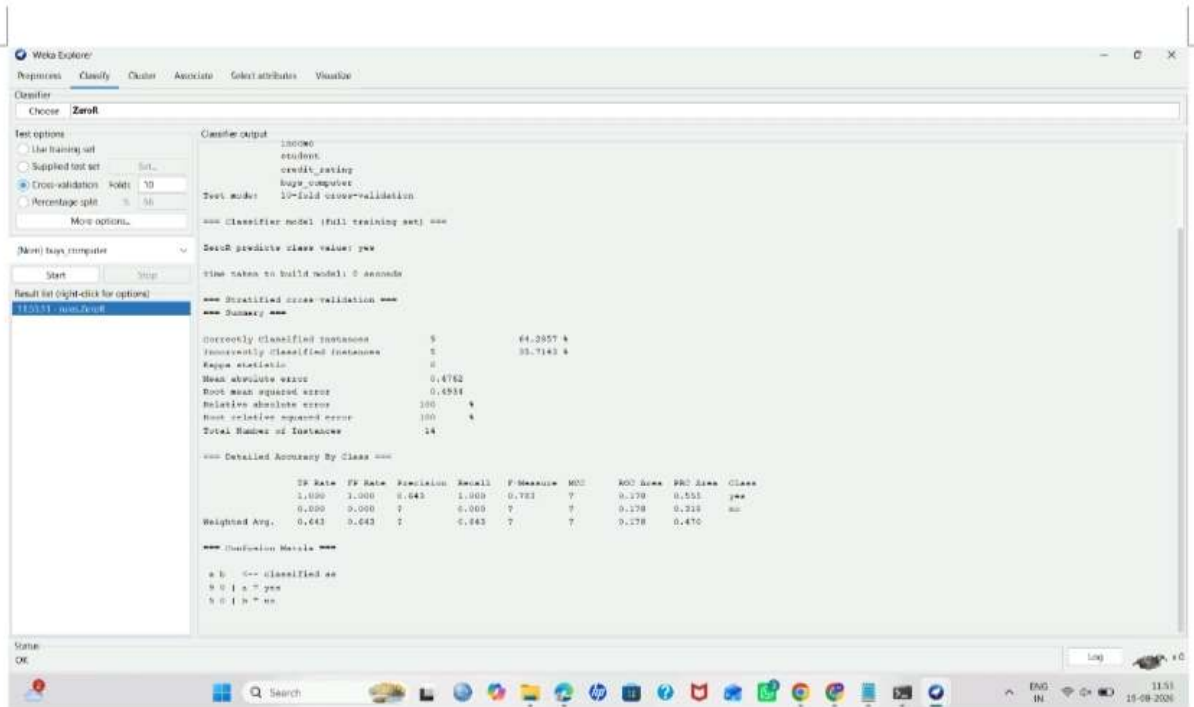
Status: OK

Log

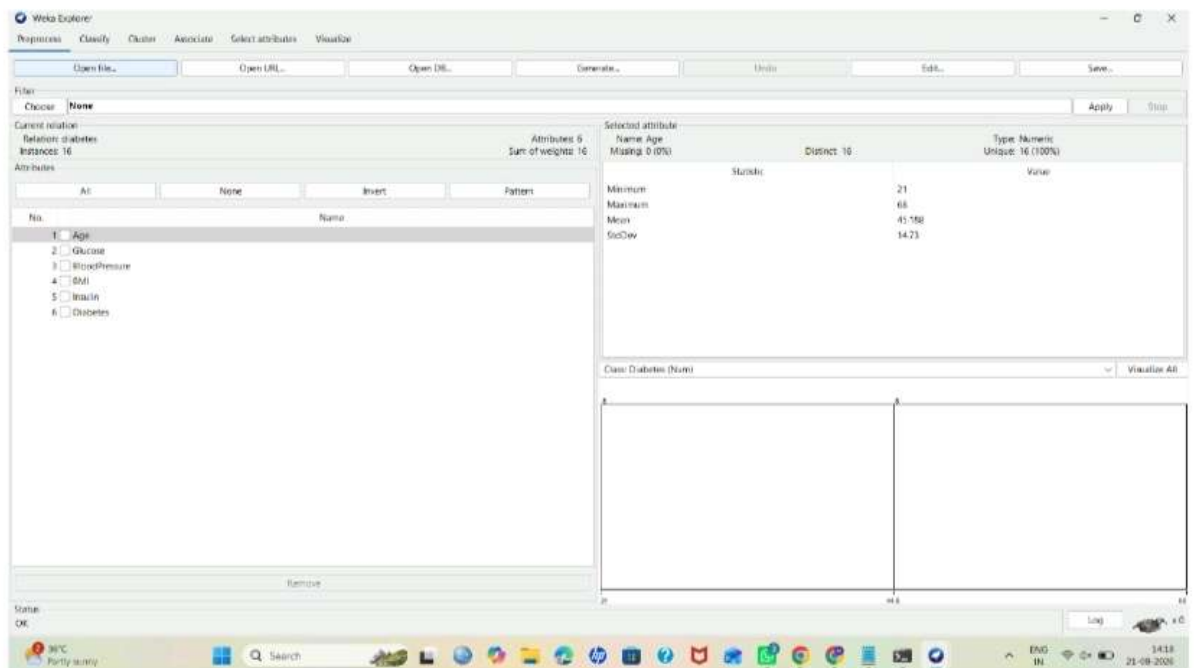


3.





4.



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: **LinearRegression** 5.0-H 1.5k 8 num-decimal-places 4

Test options:

- ☐ Use training set
- ☐ Supplied test set
- ☒ Cross-validation: Folds: 10
- ☐ Percentage split: % 50
- [More options...](#)

Start Stop

Result list (right-click for options):

147500 - LinearRegression

Classifier output:

```

=== Run information ===
Scheme: weka.classifiers.functions.LinearRegression -2 0 -R 1.0E-0 1 num-decimal-places 4
Relation: diabetes
Instances: 18
Attributes: 6
Age
Glucose
BloodPressure
sex
Insulin
Diabetes
Test mode: 10-fold cross-validation
=== Classifier model (full training set) ===

Linear Regression Model
Diabetes =

0.0341 * BloodPressure +
-2.3450

Time taken to build model: 0.05 seconds

=== Cross-validation ===
=== Summary ===

Correlation coefficient      0.8090
Mean absolute error        0.2804
Root Mean Squared Error    0.5288
Relative absolute error    71.0921 %
Root relative squared error 29.7455 %
Total Number of Instances   18
  
```

Status: OK

Log

5.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply Stop

Current relation: Relation: transactions Instances: 5

Attributes: 11 Sum of weights: 5

Selected attribute: Name: M Missing: 0 (0%)

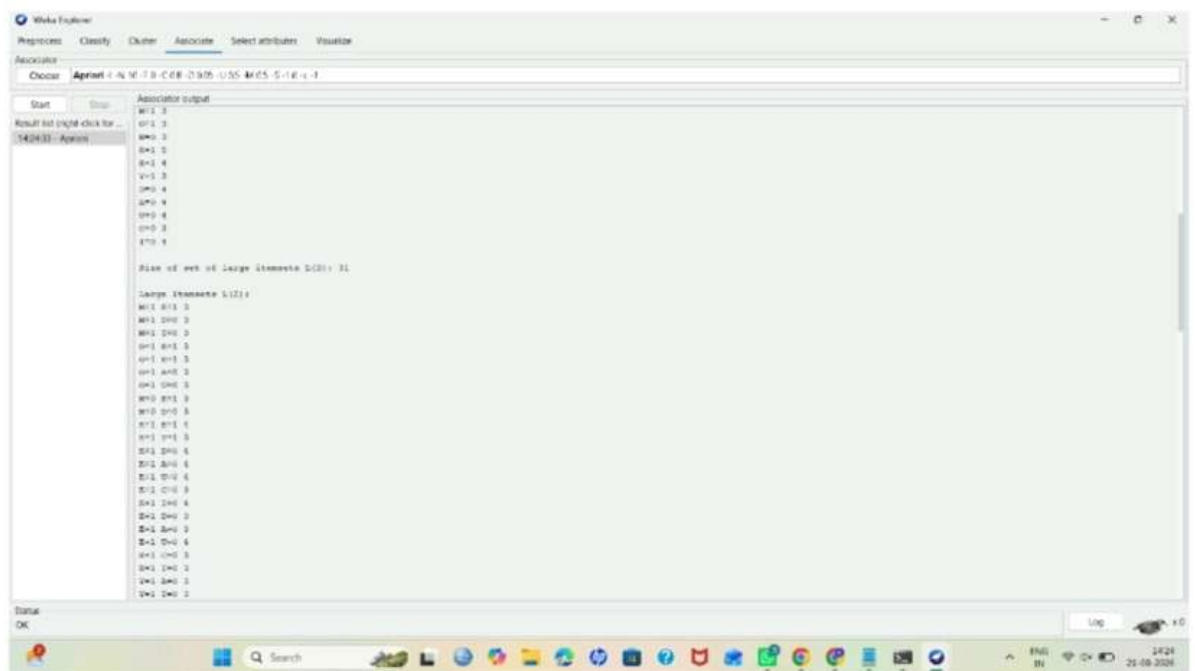
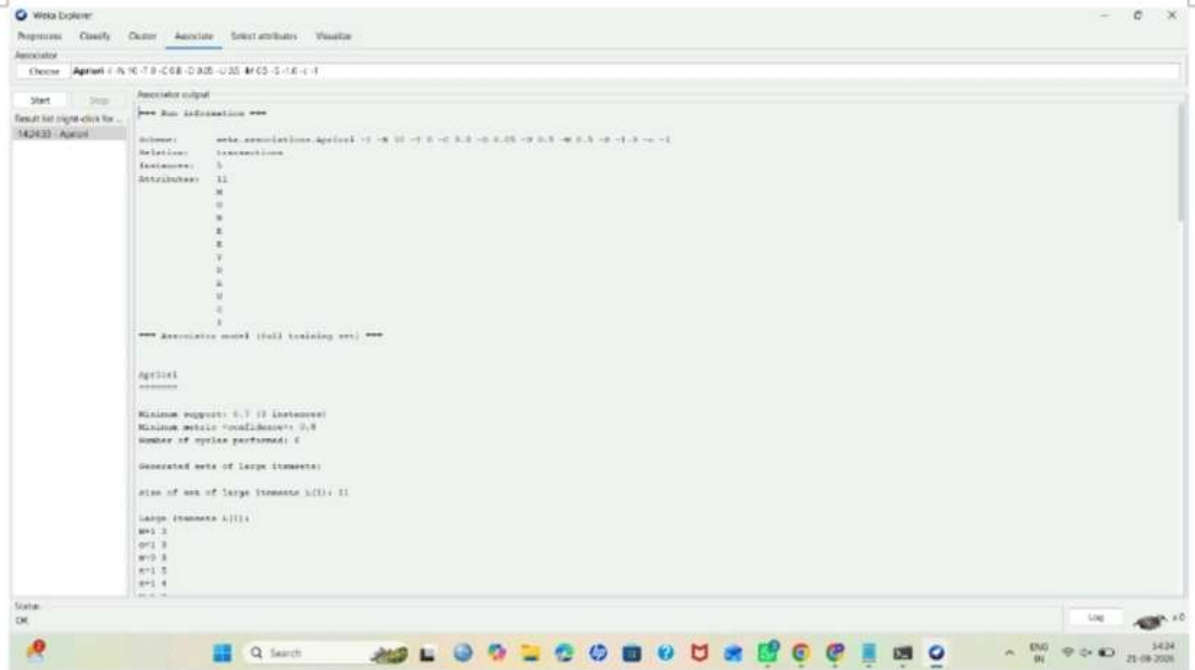
No.	Label	Distinct: 2	Count	Type: Nominal	Unique: 0 (0%)	Weight
1	0	2	2			
2	1	3	3			

Class: 1 (Name)

Visualize All

Status: OK

Log



Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: **J48 C 0.25-M 2**

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation: folds: 10
☐ Percentage split: % 66
 More options...

(More) Play
 Start Stop
 Result list (right-click for options)
 142717 - trees.j48

Classifier output:
 | Humidity = Normal: Yes (2.0)
 Outlook = Overcast: Yes (4.0)
 Outlook = Rainy
 | Windy = True: No (2.0)
 | Windy = False: Yes (2.0)

Number of leaves: 5
 Size of the tree: 8
 Time taken to build model: 0 seconds

==== Cross-validated cross-validation ====

Summary:

	Correctly classified instances	Incorrectly classified instances	Range statistic	Mean absolute error	Root mean squared error	Relative absolute error	Root relative squared error	Total Number of Instances
	7	3	-0.0425	0.4387	0.5894	27.9 %	121.2897 %	10

==== Detailed Summary by Class ====

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	AUC Area	ROC Area	Class
Yes	0.556	0.400	0.625	0.556	0.588	-0.043	0.633	0.750	Yes
No	0.400	0.444	0.233	0.400	0.304	-0.043	0.633	0.457	No
Weighted Avg.	0.500	0.544	0.521	0.500	0.500	-0.043	0.633	0.650	

==== Confusion Matrix ====

a b --> classified as
 5 4 1 a = Yes
 3 2 1 b = No

Status: OK

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier: **Logistic R 1.0E-8 M-1 num decimal places 4**

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation: folds: 10
☐ Percentage split: % 66
 More options...

(More) Play
 Start Stop
 Result list (right-click for options)
 142717 - trees.j48
 142741 - functions.logistic

Classifier output:
 Run Information:
 Schema: weka.classifiers.functions.Logistic -R 1.0E-8 -M -1 -num-decimal-places 4
 Relations: weather_prediction
 Instances: 14
 Attributes: 8
 Outlook
 Temperature
 Humidity
 Windy
 Play
 Test mode: 10-fold cross-validation

==== (Classified model) (Full training set) ====

Logistic Regression with ridge parameter of 1.0E-8
 Use infinite...

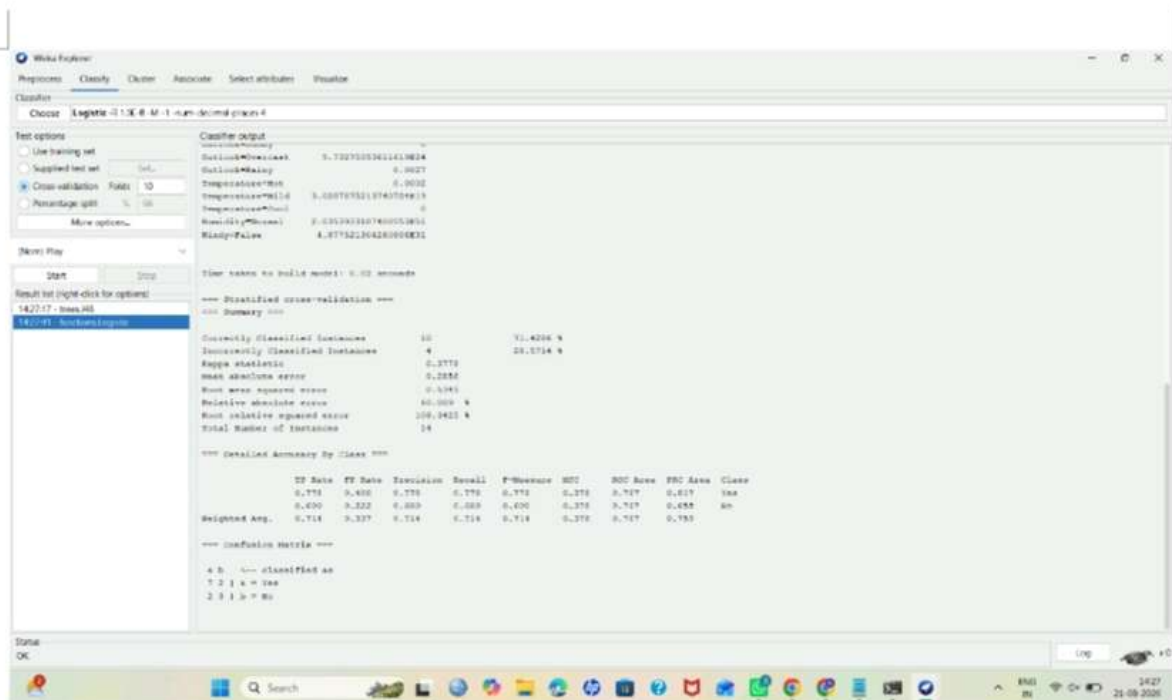
Variable Class

Variable	Class
outlook=sunny	-45.3376
outlook=overcast	87.5376
outlook=rainy	-5.9067
temperature=hot	-4.3927
temperature=mild	48.8246
temperature=cool	-48.4929
humidity=normal	118.1423
windy=false	72.9685
windy=true	-29.2032

odds ratio... Class

variable	Class
outlook=sunny	0
outlook=overcast	6.73275553611116824

Status: OK



7.

